

Team Roadmap — 12-Hour Build Plan per Person

Team Kernel Exploits · MST X Newrro Hackathon 2k26 Project: AI-mediated escrow payment gateway on MST. Read `docs/MVP.md` first for the full design. **Scoring rule change from MVP.md**: dispute scores are **integers 0–100** (percent fulfilled), not floats. See Person 2.

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0. Shared setup (everyone, hour 0–1)

What MST is to us: an EVM-compatible Layer 1 blockchain. Our Solidity contracts, viem and MetaMask work on it unchanged. The only MST-specific values are:

Setting	Value
Chain ID	<code>91562037</code>
RPC (HTTP)	<code>https://testnetrpc.mstblockchain.com</code>
RPC (WebSocket)	<code>wss://testnetrpc.mstblockchain.com</code>
Gas token	MSTC (18 decimals)
Block time	~3 s

Every person, before writing feature code:

1. `git pull`, create your branch (`chain/...`, `agent/...`, `payments/...`, `frontend/...`).
2. Add the MST testnet to MetaMask with the values above.
3. Copy `.env.example` → `.env` and fill in only what your role needs. **Never commit `.env`.**
4. Read `docs/API.md` once B1 and B2 publish it (by 0:45). Build against those shapes only. If you need a change, announce it; don't silently change it.

The five system wallets (B1 creates them, PG requests MSTC for them):

Wallet	Used by	Purpose
ORG / deployer	Backend	Deploys contracts, <code>setKyc</code> , mints MockUSD, <code>fundFor</code>
Agent	Backend (B1 calls, B2's output)	<code>proposeResolution</code> only
Arbitrator	Backend (arbitrator console)	<code>arbitrate</code> only
Demo buyer	Demo	Test user
Demo seller	Demo	Test user

Shared artifacts (single source of truth):

File	Owner	Consumers
<code>deployments.md</code>	B1	Everyone
<code>contracts/out/DealEscrow.sol/DealEscrow.json</code> (ABI)	B1	B1, PG, FE
<code>docs/API.md</code>	B1 + B2	Everyone
<code>shared/hash.ts</code> (SOW canonical hash)	B2	B2, FE
<code>shared/split.wasm</code> + <code>shared/split.ts</code>	B2	B2, FE
<code>docs/STATUS.md</code>	Everyone	Everyone, Claude Code

1. Person 1 — Backend: Chain & Data (B1)

Mission

Get our contracts live on MST, and make the backend the reliable bridge between the database and the chain. You own everything that **reads chain state** or **signs with a system key** (except the on-ramp, which is PG's).

How you use MST technology

- **MST testnet + EVM:** you deploy `MockUSD` and `DealEscrow` with Foundry against the MST RPC. Because MST is EVM-compatible, `forge` needs nothing MST-specific except the RPC URL and a funded key. Deployment is a normal contract-creation transaction, paid in MSTC.
- **MST explorer:** you verify the contracts so anyone can read the source and call `getDeal` from the browser. This is the first proof judges see.
- **MST WebSocket RPC:** your event indexer subscribes to contract events (`DealProposed`, `DealFunded`, `DisputeRaised`, `Settled`, ...) over `wss://`. MST's ~3-second blocks mean the UI updates almost live.
- **MST fast finality:** after `waitForTransactionReceipt`, the state is settled within seconds. You can safely update the database right after the receipt.
- **Gas (MSTC):** every system action (`setKyc`, `proposeResolution`, `arbitrate`) costs a small amount of MSTC. You keep the agent and arbitrator wallets funded, and you run the gas drip that gives new users enough MSTC to sign their own transactions.

Step-by-step

Hour 0–1 — Deploy the contracts (the rest of the team is blocked until this is done)

1. Set up the Foundry project:

```
cd contracts
forge init --no-git --force .
forge install foundry-rs/forge-std OpenZeppelin/openzeppelin-contracts@v5
echo '@openzeppelin/contracts=lib/openzeppelin-contracts/contracts/' > r
forge build
```

2. Generate the 5 wallets (`cast wallet new` x5). Store the private keys **only** in the backend `.env`. Post the **addresses** (never keys) in `deployments.md`.
3. Deploy:

```
export MST_RPC_URL=https://testnetrpc.mstblockchain.com
export AGENT_ADDRESS=0x... ARBITRATOR_ADDRESS=0x...
forge script script/Deploy.s.sol --rpc-url $MST_RPC_URL --private-key $OR
```

4. Verify both contracts on the MST explorer (use its verification UI, or ask mentors for the testnet verifier URL).
5. Write `deployments.md`: addresses, deploy tx hashes, explorer links. Push to `main` and tell the team.
6. With B2, write `docs/API.md`: every endpoint, plus the `Deal`, `SOW` and `DisputeScores` JSON shapes.

Hour 1–3 — Backend skeleton + chain clients + indexer

1. Express + TypeScript + SQLite (`better-sqlite3`). Tables:
 - `users(address PK, handle, kyc_level, saral_id, created_at)`
 - `deals(id PK = on-chain id, draft_id, buyer, seller, amount, status, sow_version, created_at)`
 - `drafts(id PK, buyer, seller, purpose, price, buyer_constraints, seller_points, status)`
 - `sow_versions(draft_id, version, sow_json, sow_hash, buyer_approved, seller_approved)`
 - `files(id, deal_id, kind[delivery|evidence|kyc], path, keccak)`
 - `chain_events(tx_hash, log_index, deal_id, name, args_json, block, PK(tx_hash, log_index))`
2. Chain clients (`backend/src/chain.ts`):

```
import { createPublicClient, createWalletClient, http, websocket, defineChain }
import { privateKeyToAccount } from "viem/accounts";

export const mst = defineChain({
  id: 91562037, name: "MST Testnet",
  nativeCurrency: { name: "MST", symbol: "MSTC", decimals: 18 },
  rpcUrls: { default: { http: [process.env.MST_RPC_URL!], websocket: [process.env.MST_WS_URL!] } }
});

export const pub = createPublicClient({ chain: mst, transport: websocket });
const w = (k: string) => createWalletClient({ chain: mst, transport: http });
export const org = w(process.env.ORG_KEY!);
export const agent = w(process.env.AGENT_KEY!);
export const arbitrator = w(process.env.ARBITRATOR_KEY!);
```

3. Event indexer: `pub.watchContractEvent({ address: ESCROW, abi, onLogs })`. For each log:
- Insert into `chain_events`. The primary key makes it idempotent.
 - Update `deals.status`.
 - On startup, backfill with `getContractEvents({ fromBlock: DEPLOY_BLOCK })` so a restart never loses history.

Hour 3–5 — KYC, deal reads, file hashing

1. `POST /kyc/submit`: save the uploaded file, then set `kyc_level = max(current, 1)` (mock documents pending).
2. `POST /admin/kyc/:address/approve`:
 - `org.writeContract({ functionName: "setKyc", args: [address, true] })`
→ wait for the receipt → `kyc_level = 2`.
 - Call PG's `gasDrip(address)`.
 - Return the transaction hash.
3. `GET /deals/:id`: merge the database row with `pub.readContract({ functionName: "getDeal", args: [id] })` and the list of `chain_events`. **The chain is the source of truth for status and amounts; the database is only for text and files.**
4. `POST /deals/:id/delivery` and `POST /deals/:id/evidence`: save the file and compute `keccak256(fileBytes)`. Return `{ hash }`. FE then has the **user** sign `markDelivered(id, hash)` or `raiseDispute(id, hash)`, because those are user actions, not backend actions.

Hour 5–8 — Dispute and arbitration plumbing

1. `POST /deals/:id/resolve`:
 - Load the SOW, delivery and evidence.
 - Call B2's `scoreDispute()`, which returns `{ scores, buyerBps, reasoningHash }`.
 - Call `agent.writeContract({ functionName: "proposeResolution", args: [id, buyerBps, reasoningHash] })`.
 - Store the full reasoning JSON in the database, keyed by `reasoningHash`.
2. `POST /arbitrator/deals/:id/rule`: protect it with a simple admin token. It calls `arbitrator.writeContract({ functionName: "arbitrate", args: [id, bps, keccak(rulingJson)] })`.
3. `POST /deals/:id/timeout`: calls `claimTimeout(id)` from the org wallet. Anyone can call this function, and the payout recipients are fixed by the contract.

4. Map every contract revert (`BadStatus`, `NotParty`, `TooEarly`, ...) to readable HTTP errors using viem's `decodeErrorResult`.

Hour 8–10 — Demo readiness

1. Redeploy (or deploy a second escrow) with a **120-second** `reviewPeriod` on the demo deals, so the timeout demo happens on stage.
2. `scripts/seed.ts` creates four demo deals:
 - one `Funded`, ready to deliver and release
 - one `Delivered`, ready to dispute
 - one `ResolutionProposed`
 - one `Escalated`
3. Add a reset script that re-seeds from scratch in under 2 minutes.

Handoffs

- **You give:** contract addresses, ABI, the `pub/org/agent` clients, `GET /deals/:id`, file hashes.
- **You receive:** `scoreDispute()` from B2; `gasDrip()` from PG.

Done checklist

- ☐ Contracts verified on the MST explorer; `deployments.md` Contracts verified on the MST explorer; ``deployments.md`` pushed
- ☐ Indexer survives a restart with no missing events
- ☐ Every contract revert shows up as a clear API error
- ☐ Seed/reset script works in under 2 minutes

2. Person 2 — Backend: AI Agent & SOW (B2)

Mission

Build the "brain": merge both parties' requirements into a structured, weighted SOW, and turn disputes into an explainable, deterministic, verifiable split. You also own the hashing and formula code that the frontend reuses.

How you use MST technology

- **The SOW hash is the on-chain commitment.** Your canonical `keccak256` of the SOW is what the buyer passes to `proposeDeal` and the seller to `acceptDeal`. Because the contract requires both hashes to match, your hashing function *is* the "mutually agreed beforehand" guarantee. It must be deterministic across backend and browser.
- `reasoningHash` (on-chain via `proposeResolution`) commits to the agent's full ruling: scores, formula output, model and prompt version, and the input hashes. Anyone can later recompute it and check that the ruling wasn't changed after the fact.
- **WASMify (MST's ZK-backed Web2↔chain protocol):** your split formula is the ideal candidate. It's a small, pure function. Compiled to WASM and run through WASMify, it gets a verifiable execution proof anchored to MST, which moves "trust our server" to "verify the computation." SDK access comes from mentors. **Checkpoint at hour 7:** if you don't have access by then, ship the fallback (the same WASM runs in the browser verify page).
- You never hold a key. The agent **wallet** is used by B1's endpoint. You produce the data it signs.

Step-by-step

Hour 0–1 — Schemas + canonical hashing

1. Write `shared/sow.schema.json` (JSON Schema) for the SOW:

```
{ "title": "", "buyer": "0x", "seller": "0x",
  "price": { "amount": "100.00", "token": "mUSD" },
  "deliverables": [ { "id": "D1", "description": "", "acceptanceCriteria"
  "deadline": "ISO-8601", "reviewPeriodHours": 48, "exclusions": [], "rev
```

Validation rules: `weightBps` values sum to exactly 10000; every deliverable has at least one acceptance criterion; IDs are unique.

2. `shared/hash.ts`, used by both backend and frontend:

```
import canonicalize from "canonicalize"; // RFC 8785
import { keccak256, toBytes } from "viem";
export const hashJson = (o: unknown) => keccak256(toBytes(canonicalize(o)))
```

3. Co-write `docs/API.md` with B1.

Hour 1–4 — Negotiation endpoints + SOW merge

1. `POST /deals`: the buyer creates a draft (payee address or handle, purpose, price, constraints). Status `awaiting_seller`.

2. `POST /deals/:draftId/seller-input`: the seller's points (scope, exclusions, timeline). Status `ready_to_merge`.
3. `POST /deals/:draftId/merge-sow`: calls the LLM (see §6), validates the result against the schema, and saves it as a new `sow_versions` row with `sow_hash = hashJson(sow)`.
 - If the model reports **conflicts** (for example, the buyer wants 5 days and the seller says 10), return them to the UI. Don't resolve them silently.
4. `POST /deals/:draftId/approve-sow { party, version }`: when both parties have approved the **same version**, return `{ sowHash, amount, deliverBy, reviewPeriod }`. FE uses these to have the buyer sign `proposeDeal`.
5. Any edit after approval creates a new version and resets both approvals.

Hour 4–7 — Dispute scorer + deterministic formula

1. `scoreDispute(dealId)`:
 - **Inputs**: the SOW, delivery file descriptions and hashes, the complaint text, evidence descriptions.
 - **LLM output**: `{ scores: [{ id, fulfilledPct: 0–100 integer, rationale, evidenceRefs }] }`.
 - **Validate**: every deliverable ID is present exactly once; each score is an integer from 0 to 100; no unknown IDs. On a validation failure, retry once, then error out.
2. The formula, in `shared/split.ts` and mirrored in WASM. Integer math only, so every runtime gives the same answer:

```
export function computeBuyerBps(d: { weightBps: number }[], s: { fulfilledPct: number }) {
  let bps = 0;
  for (let i = 0; i < d.length; i++) bps += Math.floor((d[i].weightBps * s[i].fulfilledPct) / 100);
  return bps; // 0..10000, buyer refund share
}
```

Example: weights 5000/3000/2000 with scores 100/50/0 give $0 + 1500 + 2000 = 3500$, so the buyer gets 35% back.

3. `reasoningHash = hashJson({ dealId, sowHash, evidenceHash, deliveryHash, scores, buyerBps, model, promptVersion })`. Store the full object in the database so the verify page can show it.
4. Return `{ scores, buyerBps, reasoningHash }` to B1's `/resolve`.

Hour 7–9 — WASM + verify + WASMify attempt

1. Port `computeBuyerBps` to Rust (about 15 lines, `wasm32-unknown-unknown`, `wasm-bindgen`) or AssemblyScript. Output `shared/split.wasm` and write a test that shows the

TypeScript and WASM versions give identical results for 20 random cases.

2. `GET /deals/:id/verify`: returns the stored reasoning object, the recomputed `buyerBps` and hash, the on-chain `reasoningHash` (via `getDeal`), and `match: true/false`.
3. **WASMify** (only if you have SDK access by hour 7):
 - Register or run `split.wasm` through WASMify with the scores as input and get the proof or receipt ID.
 - Add `wasmifyProofId` to the reasoning object before hashing, so the on-chain hash commits to the proof.
 - If WASMify isn't working by hour 9, stop. The browser verify page is the fallback.

Hour 9–10 — Tune on the demo scenarios

Run the 3 seeded disputes (clear win for the buyer, clear win for the seller, partial delivery) until the rulings look sensible and the rationales cite the evidence. Freeze the prompt version.

Handoffs

- **You give:** `shared/hash.ts`, `shared/split.ts`, `shared/split.wasm`, `scoreDispute()`, the SOW endpoints, `/verify`.
- **You receive:** file hashes and deal data from B1.

Done checklist

- ☐ The same SOW produces the same hash in Node and in the browser
- ☐ The merge never returns weights that don't sum to 10000
- ☐ The scorer rejects malformed LLM output
- ☐ TypeScript and WASM formula outputs are identical
- ☐ `/verify` returns `match: true` ``/verify`` returns ``match: true`` for every seeded ruling

3. Person 3 — Payment Gateway (PG)

Mission

Own every movement of value and every user wallet: fiat in (mock UPI) → stablecoin → escrow; settlement and refund tracking; user onboarding with SARAL; and gas for new users.

How you use MST technology

- **Stablecoin on MST:** `MockUSD` (6 decimals, same format as MST's tMUSD) is the escrowed currency. The ORG wallet is its owner, so your on-ramp mints it after a confirmed fiat payment. If mentors grant tMUSD access, the escrow is redeployed with the tMUSD address and nothing else changes.
- **`fundFor(dealId)` on the MST escrow:** the ORG never holds customer money beyond the moment of conversion. You mint → call `fundFor` → the contract pulls the tokens into escrow in the same flow. After that, only the contract's rules can move them.
- **SARAL (MST's MPC keyless onboarding):** users log in with a mobile number or Google. SARAL splits the signing key using multi-party computation, with one share tied to the user's device. This satisfies the notebook requirement of logging in through an authenticated device, and the user never sees a seed phrase. You connect the SARAL signer to viem so FE's contract calls work the same way whether the user is on SARAL or MetaMask.
- **MSTC gas:** SARAL wallets start with 0 MSTC. Your gas drip sends a small amount after KYC approval, or you use SARAL's sponsored transactions if they're supported (ask mentors).
- **BridgeKey (MST's multi-chain wallet)** and **tMUSD** are optional extras if time allows.

Step-by-step

Hour 0–2 — Mint pipeline + SARAL standalone

1. Hour 0: ask the MST mentors for (a) testnet MSTC for all 5 wallets, (b) SARAL SDK and docs, (c) WASMify access for B2, (d) tMUSD minting access.
2. Once B1 deploys, confirm the ORG wallet has approved the escrow (`Deploy.s.sol` does `approve(escrow, max)`). Check with `readContract("allowance", [org, escrow])`.
3. `backend/src/payments/mint.ts`: `org.writeContract({ address: USD, functionName: "mint", args: [orgAddress, amount] })`.
 - Amounts use 6 decimals: `parseUnits("100", 6)`.
4. SARAL standalone: run their example app **outside our codebase** first. Record exactly what the SDK returns after login: an address, a provider, or a signer function. That answer decides step 6 below.

Hour 2–4 — Mock UPI on-ramp → escrow

1. Table: `payments(id, deal_id UNIQUE, amount, status[created|paid|minted|funded|failed], mint_tx, fund_tx, created_at)`.
2. `POST /onramp/:dealId/session`: creates a payment row (`created`) and returns a fake UPI payload: amount in ₹ (use a fixed demo rate, for example ₹84 = 1 mUSD) and a QR

code or button.

3. `POST /onramp/:dealId/confirm` (the "I've paid" button in the mock UPI screen):
 1. **Idempotency**: in a DB transaction, change the status `created → paid`. If it's already past `created`, return the existing result. A double-click or retry must never mint twice.
 2. Check the on-chain state is `Accepted` via `getDeal`. Otherwise return 409.
 3. Mint → wait for the receipt → status `minted`.
 4. `org.writeContract({ address: ESCROW, functionName: "fundFor", args: [dealId] })` → wait for the receipt → status `funded`.
 5. Return `{ mintTx, fundTx }` so FE can show both explorer links.
4. Failure handling: if `fundFor` reverts after a successful mint, mark `failed` and keep the minted tokens in the ORG wallet. A retry can call `fundFor` again without re-minting.

Hour 4–6 — Settlement tracking

1. Use B1's indexed `Settled(id, toBuyer, toSeller, finalStatus)` events to fill `payouts(deal_id, to_buyer, to_seller, final_status, tx_hash)`.
2. `GET /deals/:id/payment`: returns the payment and payout records with human-readable amounts (`formatUnits(x, 6)`) and the explorer links.
3. Refund display: when `finalStatus = Refunded`, show "Refunded to buyer's wallet." For the MVP, cashing out to rupees is on the roadmap, not built.

Hour 6–9 — SARAL in the app + gas drip

1. Login flow: SARAL login → address → `POST /auth/saral { address, saralSessionProof }` → create or find the user → `kyc_level = 1` (SARAL verified the mobile number).
2. Signer adapter for FE, depending on what you found in step 4 of hour 0–2:
 - **SDK gives an EIP-1193 provider**: `createWalletClient({ chain: mst, transport: custom(saralProvider), account: address })`. Done; FE uses it like MetaMask.
 - **SDK gives only `signTransaction`**: build the transaction with viem (`prepareTransactionRequest`), sign it through SARAL, then send it with `pub.sendRawTransaction`. Wrap this in a `sendContractTx(fn, args)` helper and give it to FE.
3. `gasDrip(address)`: if the balance is below 0.05 MSTC, send 0.1 MSTC from the ORG wallet. Limit it to once per address in the database. B1 calls it after KYC approval.
4. **Checkpoint at hour 8.5**: if a SARAL user can't sign `acceptDeal` by then, MetaMask ships as the login. Keep whatever SARAL login screen works for the pitch, and be honest that it's

partial.

Hour 9–10 — Optional extras

- tMUSD: if access was granted, B1 redeploys the escrow with the tMUSD address, and you switch `mint` to tMUSD's mint method.
- BridgeKey: spend up to 30 minutes checking whether it injects a provider or supports WalletConnect. If it works, FE adds it as a wallet option. If not, drop it.
- Otherwise, run QA on the payment flow: double-clicks, a refresh mid-payment, paying for a deal that isn't `Accepted`.

Handoffs

- **You give:** `/onramp/*`, `/deals/:id/payment`, `gasDrip()`, the SARAL login + signer adapter.
- **You receive:** the ABI and addresses from B1; the Pay and Login screens from FE.

Done checklist

- ☐ Clicking "confirm payment" 5 times funds exactly once
 - ☐ The explorer shows the mint and `DealFunded` The explorer shows the mint and `DealFunded` transactions
 - ☐ Payouts are shown correctly for release, refund, split and arbitration
 - ☐ SARAL (or MetaMask fallback) login → sign `acceptDeal` SARAL (or MetaMask fallback) login → sign `acceptDeal` works end to end
-

4. Person 4 — Frontend (FE)

Mission

Make the whole flow usable and make the blockchain *visible*. Every trust-relevant step shows its on-chain proof.

How you use MST technology

- **User-signed contract calls on MST:** buyers and sellers sign their own actions from their wallet: `proposeDeal`, `acceptDeal`, `markDelivered`, `release`, `raiseDispute`, `acceptResolution`, `escalate`, and `fund` for users who already hold stablecoins. That is the non-custodial part: the backend can't do these for them.

- **MST chain config** via viem `defineChain` (chain `91562037`). Wallets prompt the user to switch to the MST network automatically.
- **MST explorer links** on every timeline step, balance and ruling. This is where judges see "Blockchain Utility."
- **SARAL**: the login screen uses PG's adapter; the rest of your code stays wallet-agnostic.
- **Verifiability**: the verify page runs B2's `split.wasm` and `hash.ts` **in the user's browser**. It recomputes the split and `reasoningHash` and compares them to the value read from MST, so the user doesn't have to trust our server.

Step-by-step

Hour 0–1 — Scaffold

1. Next.js (App Router) + Tailwind + viem (+ wagmi if you prefer hooks).
2. `lib/chain.ts`: the MST `defineChain` (README §5.6). `lib/contracts.ts`: addresses from `deployments.md` + the ABI.
3. `lib/api.ts`: typed API client **backed by mocks** that match `docs/API.md`. Switch to the live API one endpoint at a time.
4. Helpers:

```
export const txUrl = (h: string) => `${process.env.NEXT_PUBLIC_EXPLORER}/  
export const fmtUsd = (x: bigint) => formatUnits(x, 6);
```

Hour 1–4 — Onboarding + agreement screens

1. **Login**: MetaMask connect now; PG's SARAL button added later. After connecting, call `POST /auth/*`.
2. **KYC**: file upload → status badge (Level 1 = phone verified, Level 2 = approved on-chain, with a link to the `setKyc` transaction).
3. **Dashboard**: "My deals" list with Buyer and Seller tabs, showing status chips.
4. **New deal (buyer)**: payee, purpose, price, constraints → `POST /deals`.
5. **SOW workspace**:
 - Seller view: add their points.
 - Either party: "Merge with AI" → show the SOW with deliverables and weights, plus any conflicts.
 - Approve buttons for each party.
 - When both have approved:
 - The buyer signs `proposeDeal(seller, amount, sowHash, deliverBy, reviewPeriod)`. Read the new deal ID from the `DealProposed` event in the

receipt.

- The seller signs `acceptDeal(id, sowHash)`.
 - **Before signing, recompute `hashJson(sow)` in the browser with `shared/hash.ts` and refuse to sign if it doesn't match the server's hash.**
- This shows each user is checking what they sign, not trusting the server.

Hour 4–6 — Payment + tracker

1. **Pay screen:** PG's mock UPI (amount in ₹, "Pay with UPI" button) → `/onramp/:id/confirm` → show the mint and fund transaction links.
2. **Deal tracker (the key screen):** a vertical timeline built from `chain_events`. Each step shows who did it, when, the amount or hash, and a "View on explorer ↗" link. At the top, show the escrow balance for the deal, with the explorer link.
3. **Seller:** upload the delivery → B1 returns the hash → the seller signs `markDelivered(id, hash)`.
4. **Buyer:** a "Release payment" button → `release(id)`. Show the review-window countdown.

Hour 6–8 — Dispute screens

1. **Raise dispute:** complaint text + evidence files → B1 returns `evidenceHash` → the user signs `raiseDispute(id, hash)` → then call `POST /deals/:id/resolve` (the backend triggers the agent).
2. **Resolution:**
 - A table of deliverables, each with its weight, score bar, rationale and evidence references.
 - The resulting split shown in both mUSD and percent.
 - Buttons: **Accept** (`acceptResolution`) and **Escalate** (`escalate`), plus each party's acceptance status.
3. **Arbitrator console** (admin route): list escalated deals; show the SOW, evidence and agent reasoning; a split slider; "Rule" button → the arbitrator endpoint → show the transaction link.

Hour 8–10 — Verify page + SARAL + live API

1. `/deals/:id/verify`:
 - Fetch the reasoning JSON.
 - Load `split.wasm` in the browser and recompute `buyerBps`.
 - Recompute `reasoningHash` with `shared/hash.ts`.

- Read the on-chain `reasoningHash` directly from MST via `getDeal`, not from our API.
 - Show  when all three match. If a WASMify proof ID exists, show it too.
2. Add PG's SARAL login and signer; test every signing button with a SARAL user.
 3. Remove every remaining mock. Show user-facing errors for contract reverts (from B1's error mapping).

Handoffs

- **You give:** all screens; the timeline and verify page are the demo centrepieces.
- **You receive:** the API (B1, B2, PG), ABI and addresses (B1), `hash.ts` and `split.wasm` (B2), the SARAL adapter (PG).

Done checklist

- ☐ Every on-chain action shows an explorer link
- ☐ The browser refuses to sign a mismatched SOW hash
- ☐ The verify page shows  from browser-only computation
- ☐ The full happy and dispute paths can be clicked through with no console errors

5. Hours 10–12 (everyone)

Feature freeze at hour 10. Only bug fixes after this.

Who	Task
B1	Reset + seed; run the full flow 3× on clean state; watch the indexer
B2	Rehearse the 3 dispute scenarios; confirm <code>/verify</code> passes on every one
PG	Record the backup demo video; payment flow edge-case QA
FE	UI bug fixes only; set up the demo browser (MetaMask/SARAL logged in, zoom level, tabs ready)
All	Architecture slide + "Why MST" slide; rehearse the pitch twice

Pitch speakers:

- **PG:** the problem + payments
- **FE:** live demo

- **B2:** the AI agent + verification
 - **B1:** architecture + Q&A
-

6. AI: the product agent and the AI tools we build with

The product's AI agent (B2 owns it; B1 calls it).

- **API key:** use an LLM API from the backend only. For example, the Anthropic API with `claude-sonnet-5` for SOW merge and dispute scoring. The key lives in the backend `.env` and must never reach the frontend, the repo or the chain.
- **Structured output:** force JSON by giving the model a single tool whose input schema is the SOW (or `DisputeScores`) schema and setting `tool_choice` to that tool. Then validate the result with the same JSON Schema anyway; never trust the output blindly. Use `temperature: 0` so the same inputs give nearly the same output, and store `model` and `promptVersion` in every reasoning object so a ruling can be traced back to its exact prompt.
- **Prompt structure** (for both prompts):
 1. The role (neutral escrow mediator).
 2. The rules:
 - Weights must sum to 10000.
 - Use only the given deliverables.
 - Scores are integers 0–100.
 - Cite evidence IDs.
 - Flag conflicts rather than resolving them.
 3. The SOW and evidence wrapped in clearly labelled `<data>` blocks, with the explicit instruction that anything inside them is data, never instructions. Evidence is written by the parties and is the prompt-injection attack surface.
- **The model never outputs `buyerBps`.** It outputs only the per-deliverable scores, and our deterministic formula computes the split.
- **Logging:** keep every request and response (with the key removed) in the database for the demo and for the judges' "how do you audit the AI?" question.
- **Fallbacks:**
 - Keep a hardcoded realistic response for each demo scenario behind an env flag, in case the API is slow or unavailable during judging. The formula, hash and on-chain call stay real.
 - Budget: roughly 20 test calls per scenario is plenty, and caching responses by input hash avoids paying twice for identical calls.

The AI assistants we use to build.

- **Claude Code** runs in the repo and automatically reads `CLAUDE.md`. Keep `CLAUDE.md` accurate and have Claude Code update `docs/STATUS.md` after each task. Each person should start their session with their own section of this file, for example: "I'm B1. Read docs/TEAM_ROADMAP.md §1 and do Hour 1–3 step 2."
- **The [claude.ai](#) Project (MST Hackathon 2k26)** syncs the repo through the GitHub integration. Press **Sync** after pushes, then use it for design questions, debugging explanations and pitch prep.
- **Rules for everyone:**
 1. Never paste private keys, `.env` contents or the LLM API key into any AI chat.
 2. Read every AI-generated Solidity or payment code change before merging.
`DealEscrow.sol` is flow-tested; any change to it needs B1's review and a re-run of the flows.
 3. Ask for small, specific changes (one endpoint or one screen at a time), not "build the backend."
 4. When the AI suggests an MST-specific API (SARAL, WASMify), check it against the mentor-provided docs. These SDKs aren't publicly documented, so an AI can easily invent plausible-looking function names.

7. Checkpoints & fallbacks summary

Hour	Must be true	If not
1	Contracts live + verified on MST; <code>docs/API.md</code> frozen	Everyone helps B1; nobody writes feature code
4	Happy path works via API (curl)	Cut SOW editing loops; one merge only
6	Happy path works in the UI	FE drops the dashboard polish; the tracker is the priority
7	WASMify access?	No → B2 builds the browser verify page only
8	Dispute path works in the UI	Hardcode demo scores (flagged); keep the formula + chain real
8.5	SARAL user can sign?	No → MetaMask ships; SARAL shown as partial
10	Feature freeze	—

